American Computer Science League

2025-2026 ● Contest 1: Creature Capture ● Intermediate Division

PROBLEM: A creature is hiding in a cell of a 16x16 grid. You will be given the location of a guess and how far away the creature is from the guess. Find all possible cells where the creature could be located.

Each cell in the grid is represented by a 2-digit hexadecimal string (e.g. 7D). The first digit signifies the row (0-F, from top to bottom) and the second digit indicates the column (0-F, from left to right).

How far the guess is from the creature's location is called the *distance value,* given as a hexadecimal character. The distance value is the sum of the horizontal and vertical distances. For example, a distance value of 5 could be 0 rows and 5 columns, 1 row and 4 columns, 2 rows and 3 columns, etc. The following diagram illustrates a distance value of 4 from location C6 with the numbers representing the rows and columns used for that location:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0																
1																
2																
3																
4																
5																
6																
7																
8							40									
9						31		31								
A					22				22							
B				13						13						
C			04				G				04					
D				13						13						
E					22				22							
F						31		31								

After finding all possible cells in the grid where the creature could be hiding, output a string containing the smallest cell value and the largest cell value, in that order, separated by a single space.

EXAMPLE:

Input	`C6` `4`
Output	`86 F7`
Explanation	The user's guess is location C6 and the total distance value is 4. Use values of 0 and 4, 1 and 3, 2 and 2, 3 and 1, and 4 and 0. For each of these values, find the following locations that are on the grid: ● For 0 and 4, they are CA, C2. ● For 1 and 3, they are D9, D3, B9, B3. ● For 2 and 2, they are E8, E4, A8, A4. ● For 3 and 1, they are F7, F5, 97, 95. ● For 4 and 0, it is 86. The smallest cell value is 86 and the largest cell value is F7. Output `86 F7`.

INPUT: Input is two strings. The first is a 2-digit string for the guessed location. The second is a 1-digit string for the distance value, indicating the total distance horizontally and vertically that the creature is away from the guessed location.

OUTPUT: Output a string containing the smallest cell value where the creature could be hiding and the largest cell value, separated by a single space.

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SAMPLE INPUT:

1. C6
   4
2. 3D
   5
3. 0F
   3
4. 88
   6
5. 61
   7
6. 45
   A

SAMPLE OUTPUT:

1. 86 F7
2. 0B 8D
3. 0C 3F
4. 28 E8
5. 00 D1
6. 0B E5

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TEST DATA

TEST INPUT:

1. 65
   3
2. 43
   5
3. D7
   8
4. 5B
   7
5. CA
   9
6. 00
   6

TEST OUTPUT:

1. 35 95
2. 02 93
3. 57 FD
4. 09 CB
5. 3A F4
6. 06 60

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问题：一个**生物藏身于一个** 16×16 **网格的单元格中。现给**出**一个猜测位置（生物可能所在的单元格），以及该生物实际位置与猜测位置的距离**，请找出该生物可能藏身的所有单元格。

网格**中的每个单元格由一个** 2 **位十六进制字符串（**例**如** 7D**）表示。第一位数字代表行**号**（**0-F**，从上到下），第二位数字代表列**号**（**0-F**，从左到右）。**

猜测位置与生物实际位置之间的距离称为*距离值*，以一个十六进制字符表示。该距离**值是水平距离与垂直距离之和。**例如**，距离值** 5 **代表相距** 0 **行** 5 **列、**1 **行** 4 **列、**2 **行** 3 **列，以此类推。下图展示了**与**位置**C6 相距 4 距离值**的情况，其中的数字代表了该位置所对应的行和列：**

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0																
1																
2																
3																
4																
5																
6																
7																
8							40									
9						31		31								
A					22				22							
B				13						13						
C			04				G				04					
D				13						13						
E					22				22							
F						31		31								

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在找出网格中该生物可能藏身的所有单元格后，请输出一个字符串，包含最小的单元格编号和最大的单元格编号，按此顺序排列，中间用一个空格分隔。

示例：

输入	`C6` `4`
输出	`86 F7`
解析	该用户的猜测位置是 C6，总计距离值为 4。试用以下数值对：0 和 4、1 和 3、2 和 2、3 和 1、4 和 0。每对数值在网格上的位置如下： ● 0 和 4 对应的位置是 CA, C2。 ● 1 和 3 对应的位置是 D9, D3, B9, B3。 ● 2 和 2 对应的位置是 E8, E4, A8, A4。 ● 3 和 1 对应的位置是 F7, F5, 97, 95。 ● 4 和 0 对应的位置是 86。 最小单元格的值为 86，最大单元格的值为 F7。输出 `86 F7`。

输入：输入为两个字符串。第一个字符串由两位数字组成，用来表示猜测位置。第二个字符串由一位数字组成，用来表示距离值，即生物实际位置与猜测位置之间的水平和垂直总距离。

输出：输出一个字符串，由生物可能藏身单元格的最小值和最大值组成，两者之间用一个空格分隔。

样例输入：

```
C6
4
3D
5
0F
3
88
6
61
7
45
A
```

样例输出：

```
86 F7
0B 8D
0C 3F
28 E8
00 D1
0B E5
```

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数据测试

测试输入：

```
65
3
43
5
D7
8
5B
7
CA
9
00
6
```

测试输出：

```
35 95
02 93
57 FD
09 CB
3A F4
06 60
```